

Electrostatics 1

Limitation of Coulomb's Law
 → Valid only for point charges
 → Both charges must be at rest.

→ Practical Unit of charge $1 \mu A h = 3.6 C$

→ Mass of proton / Mass of electron = 1837

→ Electrons are easy to transfer than neutrons or protons.

Properties of charge:

- Quantization of charge ($q = ne$) ($n \in \mathbb{Z}$)
- Charge is always associated with mass
 No mass $\Rightarrow$ No charge.
- For an isolated system, charge is const.
- Charge is transferable.
- Charge is invariant (independent of frame)

$$m = \frac{m_{rest}}{\sqrt{1 - v^2/c^2}}$$

(Works only if at least one body is insulated)

Method of charging:

1) Friction $\Rightarrow$ Silk cloth $\Rightarrow$ +ve charge
 Metal Rod $\Rightarrow$ +ve charge

2) Conduction $\Rightarrow$ If two equal identical charged conductors touch, charge distributes equally

$$(+Q) (+Q) \Rightarrow (+Q) (+Q) \Rightarrow (+2Q) (+2Q)$$

10:50 each

3) Induction $\Rightarrow$ Method of charge separation.
 here, $F_{attr} > F_{repulsion}$.



Important Points $\Rightarrow$

1) Repulsion is true test for electrification, not attraction.

2) Large conducting surface (S) If +Q charge placed near (S) then induced charge behaves like mirror image of +Q.
 (Mirror Method)

Coulomb's Law:

In air/vacuum

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{K q_1 q_2}{r^2} \quad \epsilon_0 = 8.85 \times 10^{-12} \quad K = 9 \times 10^9$$

- Equal & opp. forces
- Always act along line joining

In a medium

$$\epsilon_r = \frac{\epsilon_m}{\epsilon_0} \Rightarrow \epsilon_m = \epsilon_0 \cdot \epsilon_r$$

(*) $\epsilon_m = K$ (dielectric const. of med)

$$F = \frac{1}{4\pi(\epsilon_0 \epsilon_r)} \frac{q_1 q_2}{r^2} \quad (F_{net} = F_e - F_{med})$$

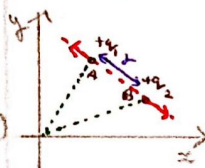
$$\Rightarrow F_{med} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \left(1 - \frac{1}{\epsilon_r}\right)$$

Imp

In other medium, F_e b/w two charges remain same. But the net force on them changes.

Vector form $\Rightarrow$

(*) Just learn to write unit vector in front of $|\vec{F}|$.



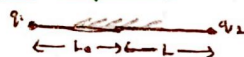
$$|\vec{F}| = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\vec{F}_A = |\vec{F}| \left(\frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} \right)$$

$$\vec{F}_B = |\vec{F}| \left(\frac{\vec{r}_{BA}}{|\vec{r}_{BA}|} \right)$$

Arranging Charges (App. of Coulomb's Law):

Case-01: No gravity $\Rightarrow$

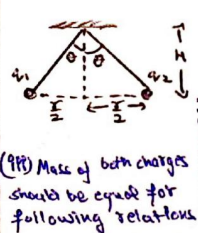


(i) Sep. b/w charges = Max. = 2L

(ii) At eqm, $\theta = \pi$

$$(iii) T = F_{elec} = \frac{K q_1 q_2}{(2L)^2}$$

Case-02: Gravity present $\Rightarrow$



(i) At eqm $\theta_1 = \theta_2 = \theta$ always because all forces on both charges are same

(ii) So by symmetry, distance of charge from normal is also same ($L/2$).

(iii) Mass of both charges should be equal for following relation

Calculate tan θ by FBD & we get

$$\frac{T}{H} = \text{const.}$$

(iv) If system is immersed in any liquid use

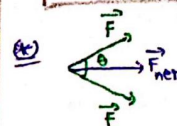
$$g_{eff} = g \left(1 - \frac{\rho_L}{\rho_B}\right)$$

$\rho_L \rightarrow$ Density of liq.
 $\rho_B \rightarrow$ Density of body (charge)

NOTE: To calc. max F_e b/w two charges, put $\frac{dF}{dx} = 0$, get mag. of charges on distance & hence get $(F_{max})_{elec}$.

Principle of Superposition of forces:

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \dots \quad (\text{vector sum})$$



$$F_{net} = 2|F| \cos\left(\frac{\theta}{2}\right)$$

(When mag. of two vectors is same)

Electric Field:

$$\vec{E} = \frac{K q}{r^2} \quad \text{Unit: } (N/C \text{ or } V/m)$$

→ For +ve charge $\Rightarrow$ radially outward
 -ve charge $\Rightarrow$ radially inward

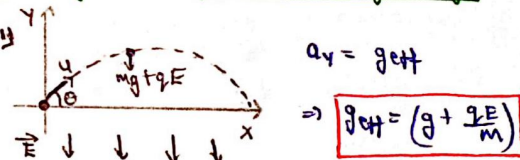


$\vec{E}$ has dirn along force acting on q_0 test charge (F_e)

Effect of electric field on test charge $\Rightarrow$

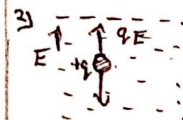
$$\vec{F} = q \vec{E}$$

Effect of electric field on moving charge $\Rightarrow$

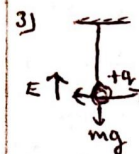


$$a_y = g_{eff}$$

$$\Rightarrow g_{eff} = \left(g + \frac{qE}{m}\right)$$



$$g_{eff} = g \left(1 - \frac{\rho_L}{\rho_B}\right) - \frac{qE}{m}$$



$$g_{eff} = \left(g - \frac{qE}{m}\right)$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{g - \frac{qE}{m}}}$$

Understanding of Test charge $\Rightarrow$

→ Test charge is infinitesimally small +ve charge (q_0)

→ Because a finite charge would induce EF on the source EF and in that case

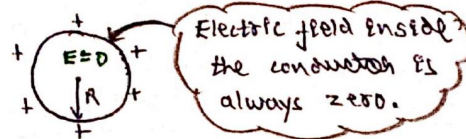
$$\vec{E} \neq \frac{\vec{F}}{q} \quad \text{Source of EF} \quad \vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$$

Hence,

$$\vec{E} = \frac{\vec{F}}{q} \quad \text{only when} \quad \vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$$

Charge Distribution on Isolated Conductors

→ In a conductor, charge is uniformly distributed on its surface only.



→ Thus, to maintain $E=0$ inside the conductor

$$\sigma \propto \frac{1}{r}$$



Electric Flux Calculation and Gauss Law

→ Electric Flux is group of field lines passing through cross section (an to field line).

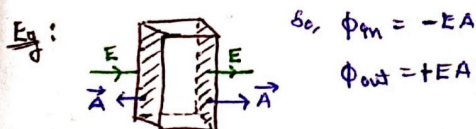
$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta$$

For non-uniform field

$$\phi = \int \vec{E} \cdot d\vec{S} = \int E \cdot ds \cos \theta$$

→ For 'flux passing through', we consider only magnitude of ϕ_{in} & ϕ_{out}

→ For 'flux emerging through', we calculate net flux with sign.



$$\text{Flux passing through} = |\phi_{in}| = |\phi_{out}| = EA$$

$$\text{Flux emerging through} = \phi_{in} + \phi_{out} = 0$$

Gauss Law

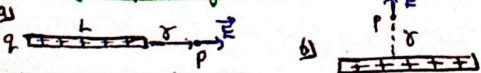
For any closed surface

$$\phi_{net out} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\epsilon_0}$$

Electric Field due to Continuous Bodies

1) Uniformly charged Rod

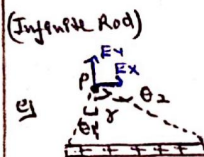


$$E_P = \frac{Kq}{L} \left(\frac{1}{r} - \frac{1}{r+L} \right)$$

$$E_P = \frac{2Kq}{\sqrt{L^2 + 4x^2}}$$

$$E_P = \frac{2K\lambda}{r} \quad (Semi-\infty wire)$$

$$E_Y = E_X = \frac{K\lambda}{r}$$



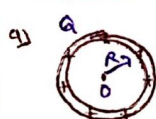
$$E_X = \frac{K\lambda}{r} (\cos \theta_2 - \cos \theta_1)$$

$$E_Y = \frac{K\lambda}{r} (\sin \theta_1 + \sin \theta_2)$$

$$E_X = \frac{K\lambda}{r} (\cos \theta_2 + \cos \theta_1)$$

$$E_Y = \frac{K\lambda}{r} (\sin \theta_1 + \sin \theta_2)$$

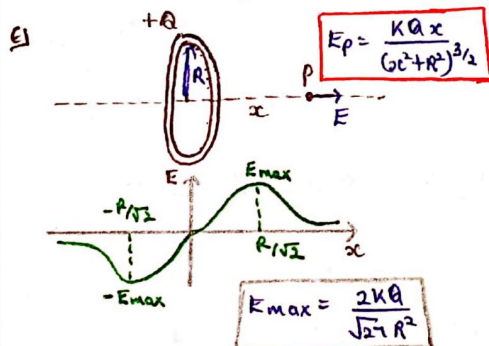
2) Uniformly charged Ring



$$E_0 = 0$$



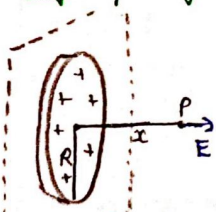
$$E_0 = \frac{Kq}{r^2} \quad [q = \frac{Q}{2\pi R}]$$



Analogy with ring:
charges kept at vertices of a regular polygon can be considered as a ring having all charge uniformly distributed on it.

$$E_P = \frac{K(q_p)x}{(x^2 + R^2)^{3/2}}$$

3) Uniformly charged Disc & Large Sheet



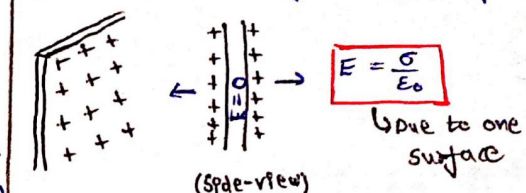
$$E_P = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

→ For a very large sheet, $R \gg x$

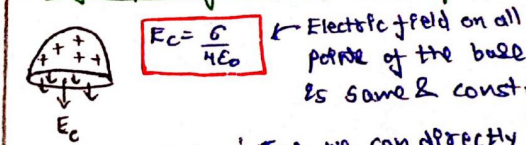
$$\Rightarrow E_P = \frac{\sigma}{2\epsilon_0}$$

1) Uniformly charged Metal Sheet

In case of metal sheet, there is no E inside the sheet & electric field gets distributed on both of its outer surfaces.



2) Uniformly charged Hemispherical Cup

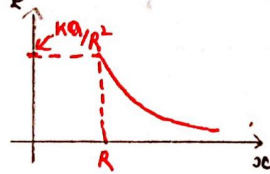
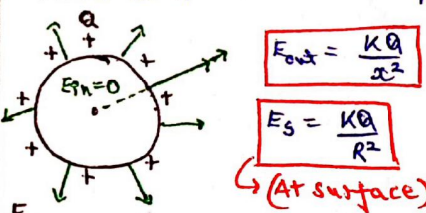


Charge is only at the surface, not in volume. True, we can directly write $\phi = E \cdot S = \frac{\sigma}{4\epsilon_0} \cdot \pi R^2$

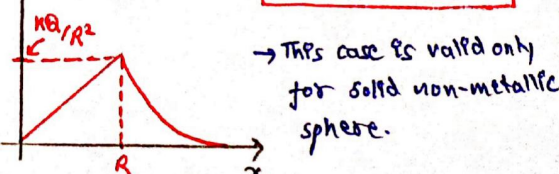
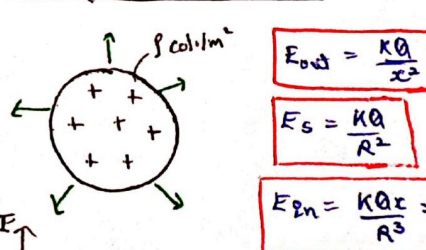
3) Uniformly charged sphere

Case-01: When charge is uniformly distributed only on the surface

→ This case is valid for solid metal / hollow metal / hollow non-metal sphere.



Case-02: When charge is uniformly distributed throughout the volume

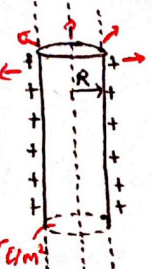


→ This case is valid only for solid non-metallic sphere.

Electrostatics - 2

1) Uniformly charged Cylinder :-

Case-01: Metallic long Cylinder $\Rightarrow$

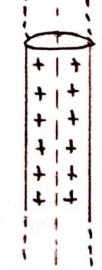


$E_{in} = 0$

$E_s = \frac{\sigma}{\epsilon_0}$

$E_{out} = \frac{\sigma R}{\epsilon_0 x}$

Case-02: Non-metallic long Cylinder $\Rightarrow$



$E_{in} = \frac{\rho x}{2\epsilon_0}$

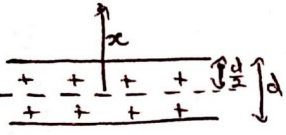
$E_s = \frac{\rho R}{2\epsilon_0}$

$E_{out} = \frac{\rho R^2}{2\epsilon_0 x}$

2) Thick charged sheet :-

$\Rightarrow$ In case of the metallic sheet, results will be same and even due to thickness, ($E_{in} = 0$).

$\Rightarrow$ But, in case of large non-metal sheet

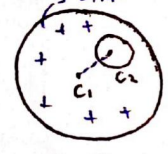


$E_{in} = \frac{\rho x}{\epsilon_0}$

$E_{out} = \frac{\rho d}{2\epsilon_0}$

(Here, x is measured from axis of sheet)

3) Spherical Cavity :-

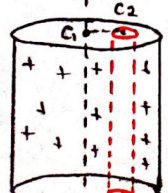


$\Rightarrow$ Inside the cavity, $\vec{E}$ is uniform everywhere.

$\Rightarrow \vec{E}$ is always along the vector joining both the centres $\vec{C_1C_2}$.

$\vec{E}_{cavity} = \frac{\rho}{3\epsilon_0} (\vec{C_1C_2})$

4) Cylindrical Cavity :-



$\Rightarrow$ Here also, $\vec{E}$ is uniform in entire cavity, and along the vector joining centres $\vec{C_1C_2}$.

$\vec{E}_{cavity} = \frac{\rho}{2\epsilon_0} (\vec{C_1C_2})$

3) Cases of Non-Uniform charge distribution

Ex (1): Calculate $\vec{E}$ due to non-metal sphere having $\rho = \rho_0 (1 + x/R)$ outside sphere.

Solⁿ

By Gauss law

$\oint \vec{E} \cdot d\vec{A} = \phi = \frac{q_{enc}}{\epsilon_0}$

$dq = \rho \cdot 4\pi x^2 \cdot dx$ (charge density) (Volume)

$\Rightarrow dq = \rho_0 (1 + \frac{x}{R}) 4\pi x^2 \cdot dx$

$q_{enc} = 4\pi \rho_0 \int_0^R (x^2 + \frac{x^3}{R}) dx$

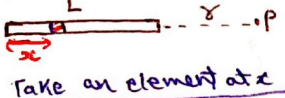
$q_{enc} = \frac{7}{3} \pi \rho_0 R^3$

So, $E \cdot 4\pi R^2 = \frac{1}{\epsilon_0} \cdot \frac{7}{3} \pi \rho_0 R^3$

$E = \frac{7 \rho_0 R}{12 \epsilon_0}$

Ans

Ex (2): Calculate $\vec{E}$ due to non-uniformly charged Rod. ($\lambda = \lambda_0 x^2$)



Solⁿ

Take an element dx

$dq_{enc} = \lambda \cdot dx = \lambda_0 x^2 \cdot dx$

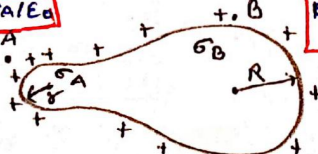
$q_{enc} = \lambda_0 \int_0^L x^2 \cdot dx = \frac{\lambda_0 L^3}{3}$

So,

$E_{net} = \int dE = \int_0^L \frac{K \cdot dq}{(L+x-x)^2}$

$E_{net} = K \lambda_0 \int_0^L \frac{x^2 \cdot dx}{(L+x-x)^2}$

12) EF near a charged Conductor :-



$E_A = \frac{\sigma A}{\epsilon_0 A} = \frac{\sigma}{\epsilon_0}$

$E_B = \frac{\sigma B}{\epsilon_0}$

① ($\sigma \propto \frac{1}{r}$) $\Rightarrow$ Only if the conductor is isolated

② net $\vec{E}$ just outside the conductor is given by $E = \frac{\sigma}{\epsilon_0}$ (σ is the charge density of nearest pt)

(Above result is valid for all situation, not necessarily for isolated conductor)

$\Rightarrow$ If above conductor is considered isolated, then

$R > r \Rightarrow \sigma_B < \sigma_A \Rightarrow E_A > E_B$

Electrostatic Potential Energy :

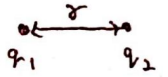
$\Rightarrow$ It is work done in assembling a system of charges.

$\Rightarrow$ At infinity, we consider $U_{\infty} = 0$

$\Rightarrow$ Since it is the work done against an electric field to bring q from ∞ ,

$U = -\vec{F}_{elec} \cdot d\vec{x} = -\int_{\infty}^r q \cdot \vec{E} \cdot d\vec{x}$

For a Two charge system $\Rightarrow$

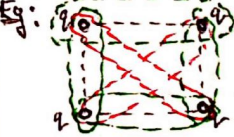


$U = \frac{K q_1 q_2}{r}$

For more than 2 charge system $\Rightarrow$

$\Rightarrow$ calculate PE of each possible pair of charge, and add them.

$\text{No. of pairs} = \frac{(n)(n-1)}{2}$ ($n \rightarrow$ No. of pt charges)

Ex:  Square of side 'a'.

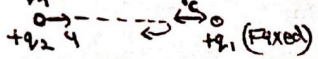
Solⁿ Total pairs = 6

4 side pairs : Distance = a

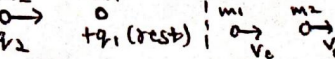
2 diagonal pairs : Distance = $a\sqrt{2}$

$(PE)_{sys} = 4 \left(\frac{K q^2}{a} \right) + 2 \left(\frac{K q^2}{a\sqrt{2}} \right)$

Closest Distance Approach :

Case-01 $\Rightarrow$  $\vec{E}_0$ (Fixed)

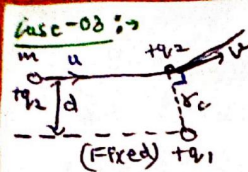
At r_c , $\frac{1}{2} m u^2 = \frac{K q_1 q_2}{r_c} \Rightarrow r_c = \dots$

Case-02 $\Rightarrow$  At r_c

$\Rightarrow$ At r_c , both charges will have same velocity.

(1) $\frac{1}{2} m u^2 = \frac{1}{2} (m_1 + m_2) v_0^2 + \frac{K q_1 q_2}{r_c}$

(2) $m_1(0) + m_2(u) = (m_1 + m_2) v_0$



When both are same charge,
path $\Rightarrow$ parabolic
When opp. charges
path $\Rightarrow$ Elliptical

$\rightarrow$ At r_c , velocity will be $\propto r_c$.

$$(1) \frac{1}{2}mv^2 = \frac{1}{2}mv^2 + \frac{kq_1q_2}{r_c}$$

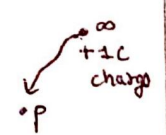
$$(2) mvd = m\delta r$$

Electric Potential :

$\rightarrow$ PE of a unit +ve charge at a point in space.

$\rightarrow$ Scalar quantity.

$$V_p = - \int_{\infty}^P \vec{E} \cdot d\vec{r}$$



$\rightarrow$ For a point charge

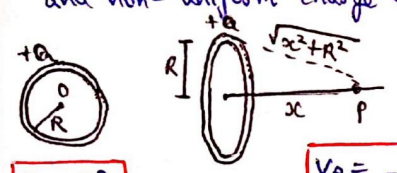


$$V_p = \frac{kq}{r} \quad (\text{note on J/C})$$

+ve charge $\Rightarrow$ +ve Potential
-ve charge $\Rightarrow$ -ve Potential

1) EP due to charged Ring :

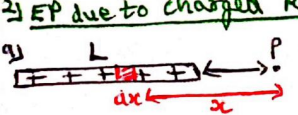
$\rightarrow$ All relations valid for both uniform and non-uniform charge distribution.



$$V_0 = \frac{kQ}{R}$$

$$V_p = \frac{kQ}{\sqrt{x^2 + R^2}}$$

2) EP due to charged Rod :

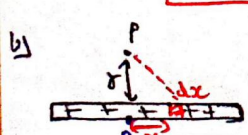


$$dV = \frac{k \cdot dq}{x}$$

$$dV = k \cdot \frac{Q}{L} \cdot dx \cdot \frac{1}{x}$$

$$\Rightarrow V_p = \frac{kQ}{L} \int_{\frac{L}{2}}^{\frac{3L}{2}} \frac{1}{x} \cdot dx$$

$$V_p = \frac{kQ}{L} \ln\left(\frac{r+L}{r}\right)$$



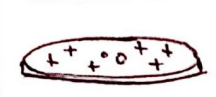
$$V_p = \int_{-L/2}^{L/2} k \cdot \frac{Q}{L} \cdot dx \cdot \frac{1}{\sqrt{r^2 + x^2}}$$

$$\Rightarrow V_p = \frac{kQ}{L} \ln\left(\frac{\sqrt{L^2 + 4r^2} + L}{\sqrt{L^2 + 4r^2} - L}\right)$$

$\rightarrow$ We can't define EP due to ∞ rod bcz due to PT, charge is present at ∞ and by convention we assume $V_{\infty} = 0$ which is contradictory.

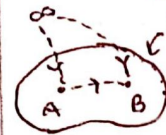
NOTE: Electric Potential can be defined only for finite bodies and not for ∞ bodies.

3) EP due to DISC :



At centre,
$$V_0 = \frac{\sigma R}{2\epsilon_0}$$

Calculation of Potential Difference :



Region of EF
 $\odot$ Since EF is a conservative field, work done does not depend on path but initial & final point.

So,

$$V_A + \left(- \int_A^B \vec{E} \cdot d\vec{r} \right) = V_B$$

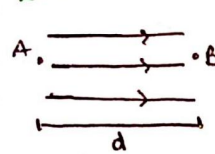
Potential for taking charge to B via A

Potential for taking charge directly to B.

(Both are equal bcz ultimately unit charge is being taken from ∞ to B)

$$\Rightarrow V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{r}$$

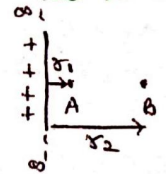
NOTE: Electric Potential uses in defn of EF.



$$V_A > V_B$$

$$V_A - V_B = Ed$$

(ii) We can calculate EP diff. wrt two pts due to ∞ bodies, if not EP.



$$V_A - V_B = \int_A^B \frac{2K\lambda}{x} \cdot dx$$

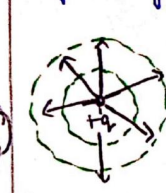
$$V_A - V_B = 2K\lambda \ln\left(\frac{x_2}{x_1}\right)$$

Equipotential Surfaces :

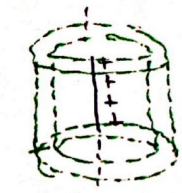
(i) EF is $\perp$ to equipotential surface at all points.

(ii) EF is either constant or zero at all points of equipotential surface.

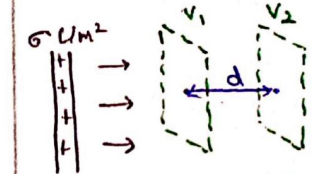
$\rightarrow$ Equipotential surface for different type of charges.



Point charge
(Concentric spheres)



Line charge
(Cylindrical surface)



Plane charge
(Parallel plane surface)

$$V_1 - V_2 = Ed = \frac{\sigma d}{2\epsilon_0}$$

Relation in EF and EP :

EF in a region is gradient of EP in space, i.e.

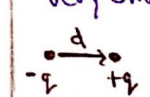
$$\vec{E} = - \nabla V \quad [\text{As } F = - \nabla V]$$

$$\vec{E} = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right] \quad (\text{3-D EF})$$

$$\vec{E} = - \frac{dV}{dr} \hat{r} \quad (\text{Unidirectional EF})$$

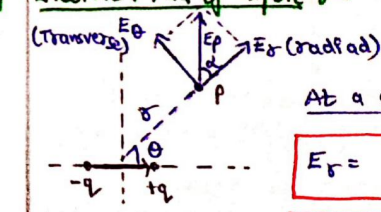
Electric Dipole :

$\rightarrow$ System of two opp. charges fixed at very small distance from each other.



Dipole moment, $\vec{P} = q\vec{d}$
(Vector quantity)

Electric Field of dipole :



At a general point

$$E_r = \frac{2Kp \cos \theta}{r^3}$$

$$E_\theta = \frac{Kp \sin \theta}{r^3}$$

$$\tan \alpha = \frac{1}{2} \tan \theta$$

$$E_p = \sqrt{E_\theta^2 + E_r^2} = \frac{Kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$$

Along the axis of dipole $\Rightarrow$

$$E_{axis} = \frac{2Kp}{r^3}$$

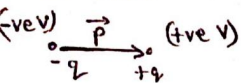
Along the equator of dipole $\Rightarrow$

$$E_{eq} = \frac{Kp}{r^3}$$

Electric Potential due to Dipole :->

→ EP along the equatorial plane is zero
bcz it is equidistant from +ve & -ve heads of dipole.

→ EP to the left is -ve & to the right is +ve.

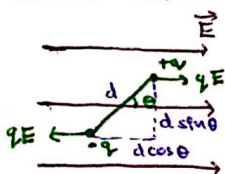


$$V = \frac{Kp \cos \theta}{r^2}$$

(θ → Angle made by position vector of general point with axis of dipole)

Dipole placed in EF :->

Case-01: Uniform Electric Field =>



$$\vec{E} = \vec{p} \times \vec{E}$$

$$C = pE \sin \theta = qE d \sin \theta$$

$$U = -\vec{p} \cdot \vec{E}$$

$$U = -pE \cos \theta$$

$$U = -qEd \cos \theta$$

Case-02: Non-Uniform Electric Field =>

In such case

$$U = -\vec{p} \cdot \vec{E}$$

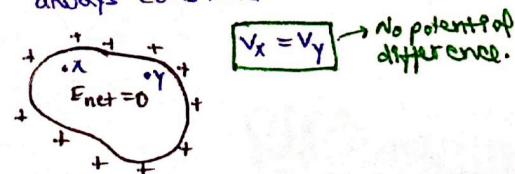
So, force on dipole is given by

$$\vec{F} = -\frac{dU}{d\vec{r}} = \left(\vec{p} \cdot \frac{d\vec{E}}{d\vec{r}} \right) \cdot \vec{r}$$

(Simply differentiate $\vec{E}$ & multiply with dipole moment)

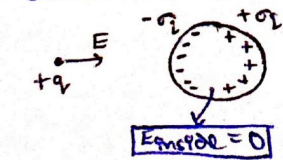
Potential Inside a Conductor :-

→ Since inside the conductor $E_{net} = 0$
So, potential inside a conductor is always constant.



$$V_x = V_y \rightarrow \text{No potential difference.}$$

→ Even under the influence of external $\vec{E}$, the charges in conductor are induced such that $\vec{E}_{in} = 0$.



Electrostatics -3

Potential due to Uniformly charged spheres :-

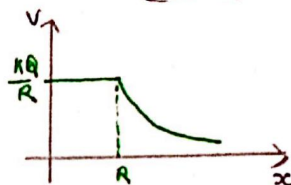
1) Conducting sphere :->



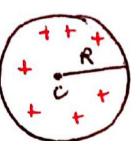
$$E_{in} = E_s = \frac{kQ}{R}$$

Inside & surface potential is same bcz ($E_{in} = 0$).

$$V_{out} = \frac{kQ}{x}$$



2) Non-Conducting sphere :->



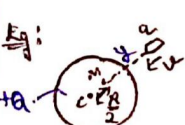
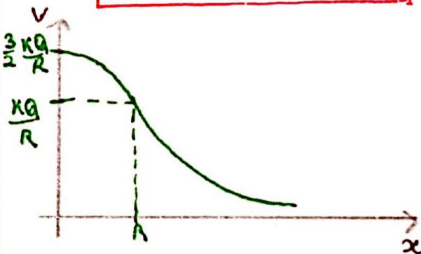
$$V_s = \frac{kQ}{R}$$

$$V_{out} = \frac{kQ}{x}$$

$$V_{in} = \frac{kQ}{2R^2} (3R^2 - x^2)$$

At centre C, $x = 0$

$$V_c = \frac{3}{2} \frac{kQ}{R} = \frac{3}{2} V_s$$



Eq: Find velocity of bullet, such that it reach till point M.

(Non-conducting uniformly charged sphere)

Sol: By Energy conservation

$$q \left(\frac{kQ}{y} \right) + \frac{1}{2} mv^2 = q \left[\frac{kQ}{2R^2} \left(3R^2 - \frac{R^2}{4} \right) \right]$$

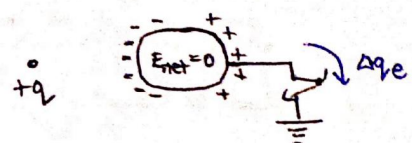
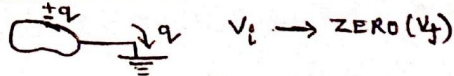
$$q \left(\frac{kQ}{y} \right) + \frac{1}{2} mv^2 = \frac{11}{8} \frac{kQ}{R}$$

$$\Rightarrow v = \sqrt{\frac{2kQq}{m} \left(\frac{11}{8R} - \frac{1}{y} \right)} \quad \text{Ans}$$

Earthing of Conductor :-

→ Earth is considered a large conducting sphere.

→ Earth makes potential of any conductor connected to it, ZERO.



→ charge will flow from conductor until it gains -ve charge big enough to have equal & opp. potential of that of external charge q.

→ So, the potential inside conductor becomes zero.

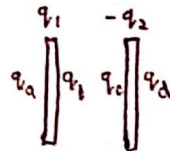
→ So, by this process, we made conductor Neutral → Negatively charged

System of Parallel Plates :-

→ In case of two parallel charged plates, the charges are present on outside & inside surfaces only, not in volume.

→ Inside charges are equal & opp. surface

→ Outside surface charges are same and mean of charges of both the plates initial.



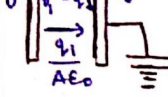
$$q_b = -q_c$$

$$q_a = q_d = \frac{q_1 + (-q_2)}{2}$$

(Always put charges with sign) ←

Earthing of the system =>

q_1, q_2 → (No significance after earthing.)



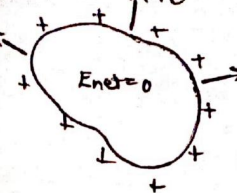
→ Outside surface charges of an earthed system are zero.

→ Inside surface charges are due to not earthed charged plate

$$q_b = -q_c \Rightarrow q_1 = -q_1$$

Electric Pressure & Energy Density of EF :-

→ Electric Pressure is defined due to surface charge of a body, in isolation.



$P_e = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2} \epsilon_0 E_{net}^2$

These expressions are valid only when body is conductor and inside EF is zero.

Energy due to EF, per unit vol.

$U_e = \frac{1}{2} \epsilon_0 E_{net}^2$

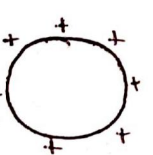
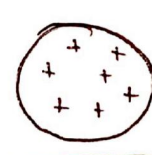
Though, this expression resembles with P_e , but this is valid everywhere. Has no limitations.

Self Energy:

→ It is defined as the work done in charging a body.

$U_s = \int \frac{1}{2} \epsilon_0 E^2 \cdot dV$

Self Energy of spheres

<u>Conducting</u>	<u>Non-conducting</u>
	
$U_s = \frac{kQ^2}{2R}$	$U_s = \frac{3}{5} \frac{kQ^2}{R}$